

Towards a Human-Centred Artificial Intelligence Maturity Model

Maria Hartikainen
Tampere University, Faculty of
Information Technology and
Communication Sciences, unit of
Computing Sciences
maria.hartikainen@tuni.fi

Kaisa Väänänen
Tampere University, Faculty of
Information Technology and
Communication Sciences, unit of
Computing Sciences
kaisa.vaananen@tuni.fi

Thomas Olsson
Tampere University, Faculty of
Information Technology and
Communication Sciences, unit of
Computing Sciences
thomas.olsson@tuni.fi

ABSTRACT

Artificial intelligence (AI) is becoming a central building block of computational systems. Following the long traditions of human-centered design, Human-Centered AI (HCAI) emphasises the importance of putting humans and various societal considerations in the centre of the development. However, the question is: how to realise HCAI when designing systems that utilise novel computational tools and require consideration of increasingly broad set of requirements, spanning from fairness and transparency to accountability and ethics? The purpose of our study is to support the AI development practices in companies in order for the humans to have AI solutions that are efficient, trustworthy, and safe. To this end, we propose a maturity model for HCAI (HCAI-MM). In this paper we present the first phase of the model development, in which the central building blocks of HCAI are specified and initial company requirements for the model's structure and content are evaluated with four AI developers.

CCS CONCEPTS

• **Computing methodologies** → Artificial intelligence; • **Human-centered computing** → Human computer interaction (HCI).

KEYWORDS

human-centered artificial intelligence, human-centered design, artificial intelligence, maturity models, company practices

ACM Reference Format:

Maria Hartikainen, Kaisa Väänänen, and Thomas Olsson. 2023. Towards a Human-Centred Artificial Intelligence Maturity Model. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3544549.3585752>

1 INTRODUCTION

Artificial intelligence (AI) is an umbrella term for several important and potent tools and technologies that have the ability to analyse vast amounts of data to solve complex problems and augment and enhance human performance [8, 10]. Recent advances in AI, like an explosion of networks and increased data processing speeds together with advances in hardware are reshaping industries and have

brought AI to a commercial level [1, 22]. The rapid development of AI technologies has resulted in AI being applied in different domains such as domestic technologies, education, sales, and finances [22, 24]. Companies are infusing AI to their internal operations as well as their products [1, 22]. In result, AI systems are becoming increasingly frequent in our everyday life, introducing positive and negative consequences for users, other people, and society [12, 14, 25]. To be competitive and benefit from the opportunities promised by AI, organisations must adapt to these new circumstances and develop new capabilities [1]. However, research show that development of AI in many sectors remains at an early stage and that AI developer companies might have low maturity AI in development [1, 14, 16]. To achieve higher maturity, companies must continually improve their capabilities and reshape their practices [7]. Reflecting the increasing importance and integration of AI in people's lives, there is a move towards human-centred artificial intelligence (HCAI), where the goal is to put the human rather than technology at the centre of AI development [25, 26]. AI solutions must not only satisfy the effectiveness and efficiency criteria familiar from human-centred design (HCD), but the utilisation of AI furthermore brings new AI-related factors, such as data bias and uncertainty, that should be addressed to reach AI applications that are trustworthy and safe for humans [13, 14, 19]. To this end, the aim of our research is to propose a maturity model (MM) that presents the building blocks of HCAI in a form that is useful and usable for AI developer companies by offering HCAI information as well as guidance and support to HCAI development. Hence, the research questions (RQs) addressed in this paper are: *RQ1 - What kind of model can help AI companies develop HCAI systems?* and *RQ2 - What are the relevant building blocks in a maturity model concerning HCAI?* We refer to the resulting model as the Human-Centred Artificial Intelligence Maturity Model (HCAI-MM). The model deals with the goal of AI developer companies meeting the requirements to develop AI that is human-centred. We are conducting this research in two phases. This work-in-progress paper reports the first phase of the study where we aim to define the relevant dimensions and the structure for the model, as well as evaluate MM usefulness as a supporting tool in HCAI development. We evaluate the preliminary model with the possible users – AI developer companies' representatives for the model to support HCAI development. After this phase, our future work will address the maturity levels and evaluate the model's usefulness and usability in real-world cases with companies.

2 BACKGROUND

Human-centered AI. Human-centered AI has been widely discussed and researched in academia in recent years (e.g., [3, 5, 8, 11, 15, 22,

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '23, April 23–28, 2023, Hamburg, Germany
© 2023 Copyright held by the owner/author(s).
ACM ISBN 978-1-4503-9422-2/23/04.
<https://doi.org/10.1145/3544549.3585752>

25, 26, 29]) and several suggestions to define the term as well as many perspectives on the topic have been presented. HCAI aims to bring efficient solutions to users' problems and provide positive and beneficial outcomes to the users, to those affected by their operation, and to society in general [18, 29]. HCAI is about understanding the impact that AI has on humans – the end-user and others [18]. Additionally, AI systems should empower both individuals and society [10, 30]. HCAI also refers to development of systems that augment rather than displace human abilities, as HCAI seeks to enhance human performance and human-AI collaboration by integrating artificial and human intelligence [9, 18, 21]. An important part of HCAI is ethical development and use of AI [14, 18, 19]. In addition, HCAI advises development of AI systems that work transparently, and allows the user to follow the functioning and the reasoning of it by being explainable and accountable [5]. HCAI seeks to preserve human control hence an important part of HCAI is to assess the need of human control over machine autonomy [25, 26]. The use of data brings the requirement of AI systems to be fair and trustworthy [10, 12, 19, 25, 27]. In addition, HCAI is also about managing the unpredictability of AI [10, 20]. Academics and governments have provided guidelines and principles for how HCAI solutions can be designed and developed (e.g., [3, 5, 8, 11, 15, 22, 25, 26, 29]). However, studies show that the requirements of HCAI are not well acknowledged currently in AI development [14, 15]. In addition, AI development practices can lean heavily on the experience, methods, and tools familiar from regular SD practices even though data and uncertainty are AI-related factors that require new capabilities, roles, and tasks [2, 15–17]. Hence, there is a need for a usable and informatic tool that can be used to support HCAI development in companies.

Maturity Models. MMs are commonly used for evaluation the maturity of processes or organisations related to a specific domain or capabilities, thus serving as benchmarks, indicating the position on the path of an evolution, and practically helping to improve development processes in the long run [4, 7]. The position is identified by assessing the competency, capability, and level of sophistication of an organization in a specific domain, based on a set of criteria that is comprehensive [7, 24]. MMs' main components are usually *dimensions* and *levels* [4, 7]. Dimensions represent specific capability areas, process areas, or design objects, which help to structure and conceptualise the field of interest [4]. Maturity levels describe criteria and characteristics that need to be satisfied to achieve a specific level of maturity, and the description of the criteria and characteristics in the next level could be used as best-practices guidelines on how to reach a higher level [4, 7, 24]. MMs are a useful tool and hundreds of them have been developed.

3 DEVELOPMENT OF HCAI-MM MODEL

The design and development of HCAI-MM is based on the maturity model development procedure proposed by [4]. This procedure includes seven steps to take that are based on eight requirements for the maturity model development. In this work-in-progress paper we consider procedure's first four steps: (i) *problem definition*, (ii) *comparison of existing models*, (iii) *determination of development strategy*, (iv) *iterative development*. This part of study covers the first iteration round. The remaining three steps, (v) *transfer*, (vi)

implementation, and (vii) *evaluation*, will be addressed in future research [4].

STEP 1 - Problem definition. Based on the procedure of [4] the development of our maturity model starts with a definition of the problem. The motivating problem of this study rises from the related academic literature and observations from recent studies of AI development practices, presented in the previous chapter. There is a need for AI development that is human-centred, however, studies show that the requirements of HCAI are not well known or acknowledged currently in AI development practices. Our objective is to support AI developers in making justified human-centred development and design decisions. Hence, the purpose of HCAI-MM is two-fold:

(i) **provide information:** the model should synthesise the building blocks of HCAI and provide this information for the AI developers in a clear and understandable way, and

(ii) **support and guide HCAI development:** the model should offer concrete guidance on how to address the requirements of HCAI. Our model provides actionable instructions, and links to existing tools.

Often MMs main goal is to continually improve IT performance with regards to its economic efficiency [5]. We are aiming the model for AI developers; however, our main goal is in the humans. We aim to develop a model to support HCAI development, so that humans would have AI solutions and products that are not only efficient and useful, but also trustworthy and safe. Nevertheless, creating AI products that are human-centered and acceptable for the humans, should be beneficial for the developer company, too, which can lead to economic efficiency.

STEP 2 – Comparison with existing models. This second procedure step should ensure that renewed research for MMs already addressing the problems and therefore developing duplicate MMs will be avoided [5]. To compare with existing human-centered AI maturity models, we conducted a literature review of HCAI related MMs. The keywords used in this search were 'human-centered AI maturity' and 'human-centered AI capability'. With this search we found one existing HCAI maturity model suggested by [29]. That model addresses the implementation of AI in the workplace while suggesting a MM of HCAI that elaborates on already existing maturity models [29]. However, this model is aimed to support the digital changes of the organisation and it does not cover AI development practices. In recent years several AI related MMs have been developed. The authors of [1] and [24] conducted a systematic literature review on existing AIMMs and based on their studies a number these models aim to specify the degree of readiness to benefit from AI in terms of improving organizational operations and processes, introducing new AI technologies, linking to the state of the company and its mindset, and estimating the performance of the organization enabled by AI technology [1, 24]. However, these models do not cover HCAI building blocks. In addition, these existing models seem to be quite ambiguous, and they do not provide concrete guidance or tools on the topic.

STEP 3 – determination of the development strategy. According to [4], the development of MMs needs to be justified according to one of the following strategies: designing a new model instead of modifying and improving an existing model. The development of a new model should transfer valid structure and content from

existing models while remaining aware of what is lacking in these models [7]. Many MMs aim to improve development processes. However, our previous study shows that AI developer companies have diverse processes and practices in AI development. Often companies aim at agile development because of the rapid progress in AI, as well as to address the uncertainty of AI projects. Therefore, we think that the best way to support HCAI development in the companies is to provide them instructions and tools that are adaptable to different processes and practices. Hence, our aim is to develop a new model that is usable and useful in diverse development practices and applicable to different life cycles. In addition, providing HCAI information to the AI developers with actionable guidance for HCAI development raises their understanding on the topic and that way can raise a company's maturity as an AI developer. Thus, the requirements for the model are:

- (i) the model is not tied to a certain process model or life cycle, but its parts can be adapted to different practices,
- (ii) the model should provide HCAI information, and offer and concrete guidance for AI developers on the topic,
- (iii) the model should support assessment of the company's current position in HCAI, and
- (iv) the model should propose practices to AI development in a way that improves the HCAI capability of the AI developer.

STEP 4 – iterative development. After the specification of the requirements for the model, the next step is design and iterative development of the model [4]. This work-in-progress paper presents the design of the preliminary model and its initial evaluation. MMs should be developed based on the theoretical foundation that is necessary to establish a well-founded development for a suitable and ready-to-use maturity model for practice and useful for other researchers [4]. In this first phase of the study, we use observations related recent HCAI literature presented in section 2 as well as material from the industry material as theoretical foundations of the model. Several major IT companies have published information and tools to support HCAI development. For the content of the model, we use this existing material because this information is aimed for AI developers, and it is presented in understandable way. We synthesise, combine, and summarise information from following sources: IBM Design for AI [18], Microsoft's Responsible AI website [19], and Google's People + AI Research (PAIR) [20]. The tools presented in the model are from these sources, too.

From the related academic literature and the industry material we recognised the following criteria for HCAI: (i) work with AI **uncertainty**: because AI systems are probabilistic, they will probably give an incorrect or unexpected output at some point. It is important to develop AI system with the knowledge that errors are integral, to understand the consequences of them and to prepare to resolve them. (ii) **user control and human-AI collaboration**: Human and AI can collaborate to enhance each other's capabilities and improve the outcome of a long, complicated process. New forms of human-AI collaborations can enhance and extend human capabilities for the good of the AI product, human, and society at large. (iii) **accountability and fairness** to ensure ethical AI development and use: AI systems should be designed in a way that respects the rule of law, human rights, democratic values, and diversity. Every person involved in the creation of AI at any step is accountable for considering the system's impact in the user and the society. In

addition, AI systems should treat all people fairly. AI systems are run by models that use data as their food, but there might be bias. Bias can manifest in any phase of the development cycle, from an unrepresentative dataset, to learned model representations, to the way in which the results are presented to the user. (iv) **explainability** and **transparency** to build trust between the human and AI: As an AI increases in capabilities and achieves a greater range of impact, its decision-making process should be explainable in terms people can understand. Explainability is key for users interacting with AI to understand the AI's conclusions and recommendations. In addition, AI systems should provide meaningful information about how the product works, including data sources and uses, privacy, and rationale behind system output, in a way that is appropriate to the context. The structure and the content of our model rise from its purpose and requirements presented previously. We are aiming to develop a multi-dimensional model, that combines dimensions and levels. Dimensions are specific capability areas or design objects structuring the field of interest [4]. In our model, we refer to these dimensions as 'building blocks'. Each building block consists of three parts. First, we provide a brief introduction on the HCAI building block and explain why it is important. Second, each building block is specified by tasks: practices or activities related to building block. Third, we provide links to online tools that can be used as further guidance. Our model will consist of three maturity levels. This work-in-progress paper presents the preliminary design of the model and specifies the first level that focuses on the base-level building blocks and practices of HCAI development. The preliminary model is presented in figure 1. The final media form and the layout of the model will be decided in step six of the model development in our future research.

4 INITIAL EVALUATION OF THE MODEL AND NEEDS ASSESSMENT

All principles and premises for the MM development, as well as usefulness, quality, and effectiveness of the artifact, must be evaluated iteratively [4]. In this initial iteration round, we (i) examined company needs related to the tools to support HCAI development in the companies, and (ii) we evaluated the model's structure and the chosen HCAI building blocks with semi-structured expert interviews.

4.1 Design of the interviews

To understand the company needs related to the HCAI development and to evaluate the preliminary model, we conducted interviews with AI developers who are potential users of the model. We had four participants from three AI and IT developer companies. See the information of the participants and the companies they present in Table 1.

We sent the preliminary model to the participants beforehand. In the first part of the interviews, we had questions of the company's needs related to HCAI development. The second part of the interviews we focused on evaluating the preliminary model and its usefulness for the AI developers. The interviews were semi-structured conduction that offered the possibility to the participants to freely give comments and ask questions. We encouraged participants to provide their honest opinion as their comments are going

EXPLAINABILITY	TRANSPARENCY	FAIRNESS	ACCOUNTABILITY	COLLABORATION AND HUMAN CONTROL	WORKING WITH AI'S UNCERTAINTY
As an AI increases in capabilities and achieves a greater range of impact, its decision-making process should be explainable in terms people can understand. Humans need to understand why AI makes the decisions it does, as trust in technology relies on understanding how it works. Explainability is key for users interacting with AI to understand the AI's conclusions and recommendations. It enables those affected by an AI system to understand the outcome and how it was arrived at	AI systems and their development should be transparent. They should provide meaningful information about how a product works, including data sources and uses, privacy, and rationale behind system output, in a way that is appropriate to the context. Transparency is important to foster general awareness and understanding of AI systems and increase acceptance and trust.	AI systems should treat all people fairly. AI systems are run by models that use data as their food. Models have to be trained with large amounts of data in order to work properly. However, in data there might be bias that can cause the model to work unfairly. Bias can manifest in any phase of the development cycle, from an unrepresentative dataset, to learned model representations, to the way in which the results are presented to the user. Errors that result from this bias can disproportionately impact some users more than others. To trust a decision made by an algorithm, it has to be fair, reliable and can be accounted for.	AI systems should be designed in a way that respects the rule of law, human rights, democratic values and diversity. Every person involved in the creation of AI at any step is accountable for considering the system's impact in the user and the society. AI designers and developers are responsible for considering AI design, development, decision processes, and outcomes.	Human and AI can collaborate to enhance each other's capabilities and improve the outcome of a long, complicated process. Some tasks, people would love for AI to handle, but there are many activities that people want to do themselves. In those latter cases, AI can help them perform the same tasks, but faster, more efficiently, or sometimes even more creatively. New forms of human-AI collaborations can enhance and extend human capabilities for the good of the AI product, human, and society at large.	AI developers have to understand an aspect that with AI there is always some level of uncertainty. Because AI systems are probabilistic, they will probably give an incorrect or unexpected output at some point. It is important to develop AI system with the knowledge that errors are integral, to understand the consequences of them and to prepare to resolve them.
TASKS: - decide which situations need explanations - explain why AI made a certain decision - explain the factors that contributed to the decision - help user to understand the AI systems capabilities and benefits rather than the technology - provide explanations that are understandable and appropriate for the user - identify if and where explanations may not be appropriate, e.g., where explanations could result in more confusion for general users, or it may negatively affect the accuracy and performance of the system - explain the aspects of data use - developers should have and maintain access to a record of an AI's decision processes and be amenable to verification of those decision processes	TASKS: - be transparent with what AI can do and how well, and explain the system's strengths and limits - clearly communicate AI limits and capabilities - present relevant information about internal processes of the AI system, to make it intuitively understandable to the user - provide appropriate transparency and control over the use of data - be transparent about where data is located and how it is used - offer information of the used training data - be transparent when your system is not certain, or cannot complete a request - increase transparency with project documenting	TASKS: - conduct testing for bias in order to make sure that the model works fairly - bias can be introduced at many stages of the AI development cycle, so test early and test often - in case of learning system, plan for continuing testing after the implementation - design and develop without intentional biases and schedule team reviews to avoid unintentional biases - make sure that the training dataset present the user groups of the developed product	TASKS: - be accountable to people by providing possibility for feedback, relevant explanations, and appeal - adhere to your company's business conduct guidelines. Also, understand national and international laws, regulations, and guidelines that your AI may have to work within - provide documentation on key decisions and ethical development throughout the AI system lifecycle make company policies clear and accessible to design and development teams from day one - understands requirements related to different AI techniques and uses - organisations have to consider the short and long-term effect of interacting with AI systems on humans, environment, and society - keep detailed records of the development process and decision making	TASKS: - understand machines impact on humans before taking actions - conclude the appropriate human direction and control - consider effective ways to human and AI to work together to enhance each other's capabilities - assess automation vs. augmentation: e.g. automate tasks that are difficult, unpleasant, or unsafe and augment tasks that people enjoy doing or they feel personally responsible of the outcome - consider the appropriate level of transparency and explainability to support human-AI collaboration and to increase trust	TASKS: - plan for the fact that your AI system will make bad predictions at some point - understand the types of errors users might encounter and have a plan for resolving them - think through the types of errors that your system could make, and their consequences - consider errors effects on the user and society, and evaluate if AI is the right technology to use - provide possibility to the users to give feedback - provide clear error messages with the reason of the error and paths forward from failure - providing manual controls when the AI fails - provide paths to contact with a customer servant in case of an error or failure
TOOLS: - IBM AI Explainability 360 – toolkit: <i>"This extensible open-source toolkit can help you comprehend how machine learning models predict labels by various means throughout the AI application lifecycle"</i> - IBM Explainability Guidelines	TOOLS: - Google: The Data Cards - Playbook: Transparent Dataset Documentation for Responsible AI: <i>"Data Cards are structured summaries of essential facts about various aspects of ML datasets needed by stakeholders across a project's lifecycle for responsible AI development"</i>	TOOLS: - IBM AI Fairness 360 Toolkit: <i>"This extensible open source toolkit can help you examine, report, and mitigate discrimination and bias in machine learning models throughout the AI application lifecycle. We invite you to use and improve it"</i> - AI Fairness Checklist - Google + TensorFlow: Responsible AI with TensorFlow -guide: <i>"A consolidated toolkit for third party developers on TensorFlow to build ML, fairness, interpretability, privacy, and security to their models"</i>	TOOLS: - Microsoft Responsible AI Standard, v2 - IBM AI Factsheet 360: <i>"Toolkit to create factsheets outlining the details of how AI service operates, how it was trained and tested, its performance metrics, fairness and robustness checks, intended uses, maintenance, and other critical details."</i> - IBM AI Privacy 360 Toolkit: <i>"The AI Privacy 360 Toolkit includes several tools to support the assessment of privacy risks of AI-based solutions, and to help them adhere to any relevant privacy requirements. Tradeoffs between privacy, accuracy, and performance can be explored at different stages in the ML lifecycle."</i>	TOOLS: - IBM model uncertainty likelihood estimation forms - Microsoft Erroranalysis: <i>"A toolkit to help to identify & Diagnose Errors, analyze and improve model accuracy. Error Analysis is a Responsible AI toolkit that enables you to get a deeper understanding of machine learning model errors"</i>	

Figure 1: – preliminary HCAI-MM model

Table 1: - Participant information

	P1	P2	P3	P4
Job title	Software consultant	CEO	Head of research	Lead AI consultant
Job tasks	AI developer	Management, project manager	AI consulting	AI consulting
Participant's experience with AI	6 years	5 years	24 years	5 years
Company type	AI development	AI development	IT development	development
Company's experience with AI	4 years	4 years	10 years	6 years

to be used to develop the model further. The interviews lasted from 54 to 76 minutes. The interviews were recorded and transcribed for analysis. In total, we had 27357 words of transcription.

4.2 Analysis and results

To analyse the content of the interview, we followed the first steps of thematic analysis procedure suggested by [6]. We coded all the transcript content, and we performed thematic grouping of the main comments brought up by the interviewees. We used Atlas.ti

for the analysis. We were concentrating especially on the comments of the model's structure and usefulness. Based on our analysis and thematic grouping, the following key results emerged:

Human-centred AI development. The term 'human-centred artificial intelligence' was familiar to all the participants even though all the participants understood it differently. To develop AI that is human-centred, P1 and P3 were hoping for information and training related to HCAI requirements. They felt that it is a management task to know the requirements and take appropriate actions to address them. P2 offered a management opinion on this by stating that the management level also needs guidance on the topic to lead the company staff. P1 and P2 hoped to have aids to educate the client about AI, as they saw the communication with the client difficult because of the lack of knowledge on AI. P3 had a HCAI related checklist used in their company, and they felt that that kind of tool is useful for HCAI development. However, they noted that the check-list tool must be short and compressed: *"too long is too long and no one will grab it. Such hundred-page instructions already exist, but it is difficult to find essential information there with just a few glances."* P2 was hoping for tool to support the initial need assessment. P4 mentioned that several guidelines and requirements related to HCAI exists, but what is lacking is a proper guidance how to really meet these high-level requirements: *"principles are often listed that there must be a fair system, for example. High-level instructions are fine, but when we are really starting to program and really have to make those solutions, that is the level where the problems start, and support and guidance ends."*

Model structure and content. HCAI-MM structure was seen clear and appropriate for the use by all the participants. The chosen HCAI building blocks seemed relevant and important, and they all were familiar with them. All the participants agreed that the content of the model was relevant to the topic and the tasks were easy to understand: *"I believe that as a big concept, at least in my opinion, these (models building blocks) are important and that you also understand why it is good to think about these things and although certainly not everything is always important to each individual case"* (P1). P4 added: *"every data scientist understands what these [the model's building blocks] mean and even though they may not know all of them beforehand, they would understand right away in this scope what they mean, so I think it is a good model. If the developer doesn't understand these, then I claim that then the challenge is more on the personal side"*. P3 commented that the model covers many important parts but still is possible to see with one glimpse. P4 appreciated how the model is in line with current HCAI related instructions and regulations. P2 noted that it should be stated clearer in the introduction why the building block is important and how acknowledging it affects the end-product. P1 and P3 hoped that there would be a possibility for more information on the building blocks without compromising the clear structure of the model. Every building block was confirmed by the participants. However, P4 stated that it is essential to include building block "data" to the model, because of data's essential role in AI development. In addition, P1 and P2 were hoping that "user and need assessment" would be acknowledged in the model, too. P2, P3, and P4 felt that it is essential to have the organisational requirements included in the model. However, everyone stated that the additional parts should be included in a way that do not compromise the clear

layout of the model. All participants felt that the tasks presented are understandable and useful. P1 suggested that providing an example of the given guidance would be more understandable and useful especially to the developers with less experience. All participants felt that the suggested tools seem to be useful in HCAI development. However, P1 and P3 hoped that there would be more information about the tools, especially what they can be used for.

Uses of the model. All participants agreed that the model could be useful especially as a check-list for the developer and to the others participating to the AI development. However, P1 and P2 doubted if the model would be comprehensive enough for those who are less familiar with the HCAI building blocks. For this, they suggested that the model could provide more information, or link to additional information without adding clutter to the model. P2 and P3 considered the model to be useful in communication with the client. They said that the client does not always understand AI or the development of it, and with the help of the model they could educate the client as well as justify the development decisions. P4 felt that the model could be used as internal training or testing material for the company employees. In addition, P2 and P3 considered that the model could be used as proof or unofficial certification of the HCAI capability.

5 DISCUSSION

This research contributes to the research field of HCI by proposing a new model that can help practitioners develop human-centred AI systems, and their capabilities in this domain. Several AI-related maturity models exist, but currently there is no actionable model for HCAI development. Hence, we were set to develop a useful tool for AI developer companies with an aim to offer them information of HCAI building blocks as well as actionable guidance and support to HCAI development. We developed preliminary HCAI-MM model that presents six HCAI building blocks, as well as actionable guidance tasks and tools support the HCAI development. The model can be used in the beginning of the development process to understand the HCAI building blocks that need to be addressed in the development, or at any phase of the development as a checklist to ensure that the HCAI buildings blocks are accounted for. The model is not tied to a certain system development process model or life cycle, but its parts can be adapted to different development practices and life cycles.

We conducted expert interviews and as an answer to the RQ1 - *What kind of model can help AI companies develop HCAI systems?* we found that companies prefer short, compressed information in a check-list form with a possibility to get more information on the topic if needed. They also hope to get targeted tasks and tools related to certain AI product or projects. For our second RQ - *What are the relevant building blocks in a maturity model concerning HCAI?* we found that there are different views of HCAI as well as suggestions of its building blocks. Thus, it was demanding to choose how to build the model so that it would cover all the necessary things but would not be too broad. Based on the evaluation, the chosen building blocks seem to be important and relevant for use. The purpose of HCAI-MM is to provide information of HCAI for the AI developers in a clear and understandable way, as well as support and guide HCAI development with actionable tasks and links to

existing tools. In the evaluation with the model's potential users, the content of the model was seen relevant and useful. However, we did not evaluate the proposed building block tasks one by one but generally, as it was not the purpose of this initial evaluation round. Hence, they will need further iteration. The model was considered useful not only to support the development of AI, but also in educating the company staff as well as the clients.

The model was seen usable as AI development support, especially as a checklist. However, we had only four participants in this first evaluation round, so in the next evaluation round we are aiming to contract more participants so that the results would be more comprehensive and thorough. In addition, all the participants were already familiar with the model's building blocks, but we do have to acknowledge readers that are less familiar with these building blocks and consider how to present enough information for them. Based on the evaluation, in the next development phase we will reflect ways of including and acknowledge 'need assessment', 'data', and 'organisational requirements' in the model without compromising the clear uncluttered layout of the model that was appreciated in the evaluation. In addition, based on the evaluation we have to consider the specifics of decisive application domains and adding a filtering option to different parts of the model, so that the developer, by filling details of their project, can directly get the information of the requirements of that certain project.

6 FUTURE WORK

The work presented in this work-in-progress paper is the first part of the development of the model. In the second part of the study, we will develop further the preliminary model based on the initial evaluation. In addition, we will consider how to support assessment of the company's current maturity and capabilities in HCAI by seeking answers to RQ3: *What kind of model can help AI companies develop their capability in HCAI?* and RQ 4 *How can HCAI maturity be expressed through specific levels?* The main tasks we will address are: (i) design of the assessment of the company's current position related to HCAI and (ii) broadening the reference pool to include further academic literature, in order to gain more knowledge on suggestions for HCAI-related actions, tools, and methods. With this information we will define different maturity levels and determine the requirements for companies to achieve higher maturity. In addition, we will evaluate the understandability and relevance of the tasks in the model individually. (iii) We will design the layout and the media form of the model, so that it would be usable for different AI developer companies. We will consider how to provide more information to those who need it, but still keep the model clear and uncluttered. In addition, we will consider how to present information on the model's purpose and use to the reader. (iv) We will compile a set of relevant tools to be utilised in HCAI development and provide targeted information of these tools, so that the reader can pick a suitable one from the list instead of going through all of them. In addition, (v) we will evaluate the next version of the model with a wider group of experts in real world AI development settings. We will also create a quantitative survey of HCAI related company needs and model's evaluation in order to get a bigger picture of HCAI-MM's usefulness in supporting the HCAI development.

ACKNOWLEDGMENTS

We thank the interviewees for their time. We are also grateful for the funders of KITE project (Human-Centered AI Solutions for the Smart City) in which this research was conducted: European Regional Development Fund, Business Tampere and University of Tampere.

REFERENCES

- [1] Alsheiabni, Sulaiman, Yen Cheung, and Chris Messom. 2019. Towards An Artificial Intelligence Maturity Model: From Science Fiction To Business Facts. In Proceedings of PACIS 2019.
- [2] Amershi, Saleema, Andrew Begel, Christian Bird, Robert DeLine, Harald Gall, Ece Kamar, Nachiappan Nagappan, Besmira Nushi, and Thomas Zimmermann. 2019. Software engineering for machine learning: A case study. In Proceedings of the 41st International Conference on Software Engineering: Software Engineering in Practise.
- [3] Auernhammer, Jan. 2020. Human-centered AI: The role of Human-centered Design Research in the development of AI. In Synergy - DRS International Conference. DOI:10.21606/drs.2020.282
- [4] Becker, Jörg, Ralf Knackstedt, and Jens Pöppelbuß. 2009. Developing maturity models for IT management. Business & Information Systems Engineering 1:3. 213-222.
- [5] van Berkel, Niels, Benjamin Tag, Jorge Goncalves, and Simo Hosio. 2020. Human-centred artificial intelligence: a contextual morality perspective. Behaviour & Information Technology. 1-17.
- [6] Braun, Virginia, and Victoria Clarke. 2012. Thematic analysis. APA handbook of research methods in psychology, Vol. 2: Research designs: Quantitative, qualitative, neuropsychological, and biological. American Psychological Association, Washington. DOI: 10.1037/13620-000
- [7] De Bruin, Tonia, Michael Rosemann, Ronald Freeze, and Uday Kaulkarni. 2005. Understanding the main phases of developing a maturity assessment model. In Australasian Conference on Information Systems (ACIS). 8-19.
- [8] Bingley, William J., Caitlin Curtis, Steven Lockey, Alina Bialkowski, Nicole Gillespie, S. Alexander Haslam, Ryan KL Ko, Niklas Steffens, Janet Wiles, and Peter Worthy. 2022. Where is the human in human-centered AI? Insights from developer priorities and user experiences. Computers in Human Behavior: 107617.
- [9] Dellermann, Dominik, Adrian Calma, Nikolaus Lipusch, Thorsten Weber, Sascha Weigel, and Philipp Ebel. 2021. The future of human- AI collaboration: a taxonomy of design knowledge for hybrid intelligence systems. arXiv preprint arXiv:2105.03354.
- [10] Dignum, Virginia. 2019. HumanE AI. Project report. Retrieved Jan 10, 2022, from <https://www.humane-ai.eu/wp-content/uploads/2019/11/D13-HumaneAI-framework-report.pdf>
- [11] Ehsan, Upol, and Mark O. Riedl. 2020. Human-Centered Explainable AI: Towards a Reflective Sociotechnical Approach. In International Conference on Human-Computer Interaction. Springer, Cham, 449-466. DOI: 10.1007/978-3-030-60117-1_33
- [12] Feuerriegel, Stefan, Mateusz Dolata, and Gerhard Schwabe. 2020. Fair AI: Challenges and Opportunities. Business & information systems engineering 62:4. 379-384. DOI: 10.1007/s12599-020-00650-3
- [13] Giacomini, Joseph. 2014. What is human centred design? The Design Journal 17, 4 (2014): 606-623. DOI:10.2752/175630614X14056185480186
- [14] Hagendorff, Thilo. 2020. The ethics of AI ethics: An evaluation of guidelines. Minds and Machines 30, 1 (2020). 99-120. DOI:10.1007/s11023-020-09517-8
- [15] Hartikainen, Maria, Kaisa Väänänen, Anu Lehtiö, Saara Ala-Luopa, and Thomas Olsson. 2022. Human-Centered AI Design in Reality: A Study of Developer Companies' Practices: A study of Developer Companies' Practices. In Nordic Human-Computer Interaction Conference. 1-11.
- [16] Hesenius, Marc, Nils Schwenzfeier, Ole Meyer, Wilhelm Koop, and Volker Gruhn. 2019. Towards a software engineering process for developing data-driven applications. In Proceedings of 7th International Workshop on Realizing Artificial Intelligence Synergies in Software Engineering. ACM, Montreal, 35-41. DOI: 10.1109/RAISE.2019.00014
- [17] Hill, Charles, Bellamy, Rachel, Erickson, Thomas, and Burnett, Margaret. 2016. Trials and Tribulations of Developers of Intelligent Systems: A Field Study. In 2016 IEEE Symposium on Visual Languages and Human-Centric Computing. 162-170. DOI: 10.1109/VLHCC.2016.7739680
- [18] IBM Design for AI. Retrieved Jan 2, 2022 from <https://www.ibm.com/design/ai/>
- [19] Microsoft Responsible AI. Retrieved Jan 2, 2022 from <https://www.microsoft.com/en-us/ai/responsible-ai-resources>
- [20] PAIR with Google. People + AI Research. Retrieved Jan 3, 2022 from <https://pair.withgoogle.com>.
- [21] Passi, Samir and Steven J. Jackson. 2018. Trust in Data Science: Collaboration, Translation, and Accountability in Corporate Data Science Projects. Proceedings

- of the ACM on Human-Computer Interaction, 2: 1–28. DOI: 10.1145/3274405
- [22] Perrault, R., Shoham, Y., Brynjolfsson, E., Clark, J., Etchemendy, J., Grosz, B., Lyons, T., Manyika, J., Niebles, J. C. and S. Mishra. 2019. The AI Index 2019 Annual Report. AI Index Steering Committee. Human-Centered AI Institute. Stanford University.
 - [23] Pöppelbuß, Jens, and Maximilian Röglinger. 2011. What makes a useful maturity model? A framework of general design principles for maturity models and its demonstration in business process management. In ECIS 2011 proceedings.
 - [24] Sadiq, Raghad Baker, Nurhizam Safie, Abdul Hadi Abd Rahman, and Shidrokh Goudarzi. 2021. Artificial intelligence maturity model: a systematic literature review. *PeerJ Computer Science* 7: e661.
 - [25] Shneiderman, Ben. 2020. Bridging the gap between ethics and practise: Guidelines for reliable, safe, and trustworthy Human-Centered AI systems. *ACM Transactions on Interactive Intelligent Systems*, 10: 4: 1–31. DOI: 10.1145/3419764
 - [26] Shneiderman, Ben. 2020. Human-Centered Artificial Intelligence: Reliable, Safe & Trustworthy. *International Journal of Human-Computer Interaction* 36:6: 495–504. DOI:10.1080/10447318.2020.1741118
 - [27] Wan, Zhiyuan, Xin Xia, David Lo, and Gail C. Murphy. 2019. How Does Machine Learning Change Software Development Practices? *IEEE Transactions on Software Engineering*, 47:9: 1857–1871. DOI: 10.1109/TSE.2019.2937083
 - [28] Wickramasinghe, Chathurika, S., Daniel L. Marino, Javier Grandio, and Milos Manic. 2020. Trustworthy AI Development Guidelines for Human System Interaction. In *International Conference on Human System Interaction*. 130–136. DOI: 10.1109/HSI49210.2020.9142644
 - [29] Wilkens, Uta, Valentin Langholf, Greta Ontrup, and Annette Kluge. 2021. Towards a Maturity Model of Human Centered AI—A Reference for AI Implementation at the Workplace. *Forschungsseminar der Wissenschaftlichen Gesellschaft für Arbeits- und Betriebsorganisation (WGAB)*. DOI:10.30844/wgab_2021_11
 - [30] Xu, Wei. 2019. Toward Human-Centered AI: A Perspective from Human-Computer Interaction. *Interactions* 26:4: 42–46. DOI: 10.1145/3328485